

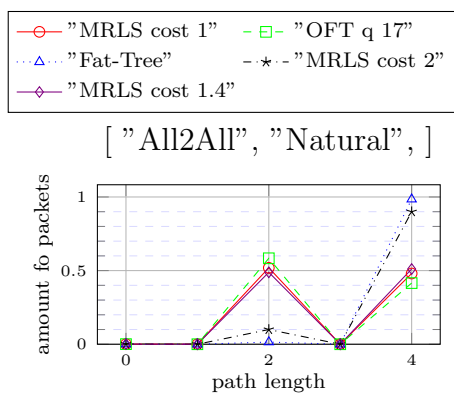


Figure 1: X: ["All2All", "Natural",]

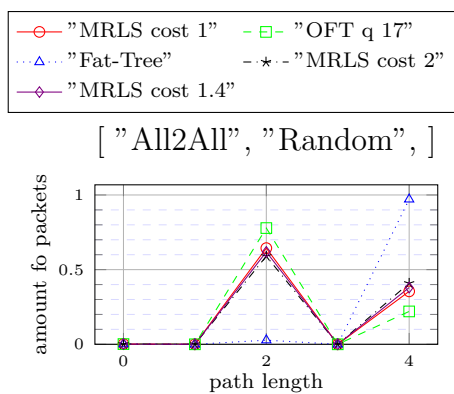


Figure 2: X: ["All2All", "Random",]

The following versions used in the simulations.

- heads/alex-dev-defe967c1db10e3b873fe074f274b6bdc9c42bb8(0.6.3)
- heads/master-dirty-65e4f7db651d154c0abb3e3d0a691be7d6d92d82(0.6.3)
- heads/master-ce88a576295ce68f40736599a9e3e04f24f38801(0.6.3)

```

Configuration {
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 1000000000000000,
  general_frequency_divisor: 2,
  topology: topo![
    MultiStage {
      stages: [
        ExplicitStageFile { filename: "../MRLS_614_18_307_36" },
        servers_per_leaf: 18,
        legend_name: "MRLS cost 1",
        OFT { height: 1, prime: 17, servers_per_leaf: 18, legend_name: "OFT q 17" },
      ]
    }
    MultiStage {
      stages: [
        Fat { bottom_factor: 18, top_factor: 18 },
        Fat { bottom_factor: 36, top_factor: 18 },
        servers_per_leaf: 18,
        legend_name: "Fat-Tree",
      ]
    }
    MultiStage {
      stages: [
        ExplicitStageFile { filename: "../MRLS_972_24_648_36" },
        servers_per_leaf: 18,
        legend_name: "MRLS cost 2",
      ]
    }
    MultiStage {
      stages: [
        ExplicitStageFile { filename: "../MRLS_756_21_441_36" },
        servers_per_leaf: 18,
        legend_name: "MRLS cost 1.4" },
      ]
    }
  ]
  traffic: topo![
    TrafficMap {
      tasks: 11052,
      map: ![
        Identity { legend_name: "Natural" },
        RandomPermutation { legend_name: "Random" },
      ],
      application: ![
        All2All { tasks: 11052, data_size: 176832 },
      ],
      legend_name: "All2All",
    }
    TrafficMap {
      tasks: 11052,
      map: ![
        Identity { legend_name: "Natural" },
        RandomPermutation { legend_name: "Random" },
      ],
      application: ![
        All2All { tasks: 11052, data_size: 176832 },
      ],
      legend_name: "All2All",
    }
    TrafficMap {
      tasks: 11664,
      map: ![
        Composition {
          patterns: [
            Identity { legend_name: "Natural" },
            CartesianEmbedding {
              source_sides: [ 11052 ],
              destination_sides: [ 11664 ] },
            middle_sizes: [ 11052, 11664 ],
            legend_name: "Natural" },
          Composition {
            patterns: [
              RandomPermutation { legend_name: "Random" },
              CartesianEmbedding {
                source_sides: [ 11052 ],
                destination_sides: [ 11664 ] },
              middle_sizes: [ 11052, 11664 ],
              legend_name: "Random" },
            ],
          application: ![
            All2All { tasks: 11052, data_size: 176832 },
          ],
          legend_name: "All2All",
        }
      ],
      legend_name: "All2All",
    }
    TrafficMap {
      tasks: 11664,
      map: ![
        Composition {
          patterns: [
            Identity { legend_name: "Natural" },
            CartesianEmbedding {
              source_sides: [ 11052 ],
              destination_sides: [ 11664 ] },
            middle_sizes: [ 11052, 11664 ],
            legend_name: "Natural" },
          Composition {
            patterns: [
              RandomPermutation { legend_name: "Random" },
              CartesianEmbedding {
                source_sides: [ 11052 ],
                destination_sides: [ 11664 ] },
              middle_sizes: [ 11052, 11664 ],
              legend_name: "Random" },
            ],
          application: ![
            All2All { tasks: 11052, data_size: 176832 },
          ],
          legend_name: "All2All",
        }
      ],
      legend_name: "All2All",
    }
    TrafficMap {
      tasks: 11340,
      map: ![
        Composition {
          patterns: [
            Identity { legend_name: "Natural" },
            CartesianEmbedding {

```

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    source_sides: [ 11052 ],
    destination_sides: [ 11340 ]]],
    middle_sizes: [ 11052, 11340 ],
    legend_name: "Natural"],
Composition{
  patterns: [
    RandomPermutation{ legend_name: "Random" },
    CartesianEmbedding{
      source_sides: [ 11052 ],
      destination_sides: [ 11340 ]]],
    middle_sizes: [ 11052, 11340 ],
    legend_name: "Random"]],
  application: ![
    All2All{ tasks: 11052, data_size: 176832 }],
  legend_name: "All2All"}],
maximum_packet_size: 16,
router: InputOutput{
  virtual_channels: topo![ 4, 4, 2, 4, 4 ],
  virtual_channel_policies: topo![
    [
      MapHop{
        hop_to_policy: [
          ArgumentVC{
            allowed: [ 0 ]},
          ArgumentVC{
            allowed: [ 0 ]},
          ArgumentVC{
            allowed: [ 1 ]},
          ArgumentVC{
            allowed: [ 1 ]},
          ArgumentVC{
            allowed: [ 2 ]},
          ArgumentVC{
            allowed: [ 2 ]},
          ArgumentVC{
            allowed: [ 3 ]},
          ArgumentVC{
            allowed: [ 3 ]}],
          VecLabel{
            label_vector: [ 0, 64, 80 ]},
          OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
          OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
          LowestLabel,
          EnforceFlowControl,
          Random],
        [
          MapHop{
            hop_to_policy: [
              ArgumentVC{
                allowed: [ 0 ]},
              ArgumentVC{
                allowed: [ 0 ]},
              ArgumentVC{
                allowed: [ 1 ]},
              ArgumentVC{
                allowed: [ 1 ]},
              ArgumentVC{
                allowed: [ 2 ]},
              ArgumentVC{
                allowed: [ 2 ]},
              ArgumentVC{
                allowed: [ 3 ]},
              ArgumentVC{
                allowed: [ 3 ]}],
              VecLabel{
                label_vector: [ 0, 64, 80 ]},
              OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
              OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
              LowestLabel,
              EnforceFlowControl,
              Random],
            [
              OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
              OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
              LowestLabel,
              EnforceFlowControl,
              Random],
            [
              MapHop{
                hop_to_policy: [
                  ArgumentVC{
                    allowed: [ 0 ]},
                  ArgumentVC{
                    allowed: [ 0 ]},
                  ArgumentVC{
                    allowed: [ 1 ]},
                  ArgumentVC{
                    allowed: [ 1 ]},
                  ArgumentVC{
                    allowed: [ 2 ]},
                  ArgumentVC{
                    allowed: [ 2 ]},
                  ArgumentVC{
                    allowed: [ 3 ]},
                  ArgumentVC{
                    allowed: [ 3 ]}],
                  VecLabel{
                    label_vector: [ 0, 64, 80 ]},
                  OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },

```

```

OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
LowestLabel,
EnforceFlowControl,
Random],
[
MapHop{
hop_to_policy: [
ArgumentVC{
allowed: [ 0 ]},
ArgumentVC{
allowed: [ 0 ]},
ArgumentVC{
allowed: [ 1 ]},
ArgumentVC{
allowed: [ 1 ]},
ArgumentVC{
allowed: [ 2 ]},
ArgumentVC{
allowed: [ 2 ]},
ArgumentVC{
allowed: [ 3 ]},
ArgumentVC{
allowed: [ 3 ]}],
VecLabel{
label_vector: [ 0, 64, 80 ]},
OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
LowestLabel,
EnforceFlowControl,
Random]],
allocator: Random,
buffer_size: 128,
bubble: false,
flit_size: 16,
intransit_priority: false,
allow_request_busy_port: true,
output_buffer_size: 64,
crossbar_frequency_divisor: 1,
crossbar_delay: 2},
routing: topo![
Polarized{ include_labels: true, strong: false, legend_name: "Polarized" },
Polarized{ include_labels: true, strong: false, legend_name: "Polarized" },
UpDown{ legend_name: "minimal routing" },
Polarized{ include_labels: true, strong: false, legend_name: "Polarized" },
Polarized{ include_labels: true, strong: false, legend_name: "Polarized" }],
link_classes: [
LinkClass{ delay: 2 },
LinkClass{ delay: 2 },
LinkClass{ delay: 2 }],
launch_configurations: [
Slurm{
job_pack_size: 1,
time: "3-00:00:00",
sbatch_args: [ "--account=uc95", "--qos=gp_resa", "--cpus-per-task=4", "--ntasks=1" ]}]}
```